

wire feeder item table

No.	Item	Single-material Part / Assembly	Function	Composition	Materials	Need Rationale Source / Page / Evidence	Material Source / Page / Evidence
WF-1	Wire spool	Single part	Store feedstock wire before feeding.	Spool body, wound feedstock wire, spool bore.	Feedstock wire material belongs to the material inventory; spool body may be supplier-dependent plastic, steel, or aluminum.	Miller S-74S/S-74D owner's manual, an industrial wire-feeder operation and parts manual; p.17: "Install wire spool." NASA EBF ³ space-environment paper confirms the flight-weight feeder used standard welding-wire spools; p.3: "standard 100 mm diameter welding wire spools."	Supplier-dependent. Do not lock spool body material from generic wire-feeder manuals.
WF-2	Spool hub	Single part	Support the spool bore and allow the spool to rotate on the feeder.	Hub body.	Aluminum alloy or steel; for a lunar/vacuum version, prefer aluminum alloy or stainless steel.	Miller S-74S/S-74D parts list; p.28: "HUB & SPINDLE ASSEMBLY" and "HUB, spool."	Miller confirms the part but does not lock the alloy. Material is an engineering selection.
WF-3	Spool support shaft	Single part	Provide the rotation axis for the spool hub.	Shaft body.	Stainless steel or plated steel; for lunar/vacuum use, avoid corrosion-prone plain steel.	Miller S-74S/S-74D parts list; p.28: "SHAFT, support spool."	Material is an engineering selection; source confirms the part, not the alloy.
WF-4	Spool retaining ring	Single part	Prevent the spool or hub from sliding off the support shaft.	Retaining ring.	Spring steel or stainless retaining ring.	Miller S-74S/S-74D parts list; p.28: "RING, retaining spool."	Material is an engineering selection; final material should come from a retaining-ring supplier.
WF-5	Spool brake washer	Single part	Add friction so the spool does not overrun when wire feeding stops.	Brake washer.	Steel brake washer for conventional systems; vacuum-compatible friction material if a nonmetallic friction element is needed.	Miller S-74S/S-74D parts list; p.28: "WASHER, brake stl."	Miller explicitly identifies a steel brake washer. For vacuum/lunar use, any nonmetallic friction material needs separate compatibility testing.
WF-6	Spool tension nut	Single part	Adjust spool brake pressure so wire remains taut after feed stops.	Self-locking nut or adjustment nut.	Stainless steel nut, plated steel nut, or aluminum knob with steel insert.	Miller S-74S/S-74D threading procedure; p.17: "Adjust tension nut so wire is taut when wire feed stops."	Material is an engineering selection; final part should be chosen by vacuum compatibility and maintenance method.
WF-7	Gearmotor	Assembly	Rotate the drive-roll mechanism at the commanded feed speed.	Motor, gearbox, output shaft, brush set if brushed, motor leads.	Vacuum-compatible stepper/servo or DC motor; stainless/aluminum mount; Kapton/PTFE/PEEK insulated wiring for lunar/vacuum use.	Miller S-74S/S-74D drive-assembly parts list; p.31: "MOTOR, gear."	Miller confirms the gearmotor, but not vacuum material details. Final material must come from a vacuum-rated motor datasheet.
WF-8	Motor clamp insulator	Single part	Electrically/mechanically isolate or seat the motor clamp region.	Insulator body.	Ceramic, glass-filled polymer, or vacuum-compatible polymer; for lunar/vacuum use,	Miller S-74S/S-74D main assembly parts list; p.28: "INSULATOR, motor clamp."	Material is not specified by Miller; final material requires vendor data and vacuum

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					prefer ceramic or qualified low-outgassing polymer.		compatibility review.
WF-9	Motor base clamp	Single part	Hold the motor or motor base in the feeder frame.	Clamp body.	Aluminum alloy or stainless steel.	Miller S-74S/S-74D main assembly parts list; p.28: "CLAMP, motor base."	Material is an engineering selection; source confirms component existence.
WF-10	Drive roll carrier	Single part	Hold each drive roll in the correct position relative to the wire path.	Carrier body, bearing features, fastener features.	Steel, stainless steel, or aluminum; stainless is preferred for vacuum/hot-zone robustness.	Miller S-74S/S-74D drive-assembly parts list; p.31: "CARRIER, drive roll."	Material is an engineering selection; source confirms the component, not the alloy.
WF-11	Drive roll carrier spacer	Single part	Set spacing/alignment of the drive-roll carrier.	Spacer body.	Stainless steel, plated steel, or aluminum alloy.	Miller S-74S/S-74D drive-assembly parts list; p.31: "SPACER, carrier drive roll."	Material is engineering-selected based on stiffness, wear, and vacuum compatibility.
WF-12	Drive roll	Single part	Grip the wire and convert motor torque into wire feed motion.	Grooved roll body.	Hardened steel or stainless steel; polished U-groove roll for aluminum wire; V-groove roll for solid hard wire.	Miller S-74S/S-74D parts list includes drive-roll kits; p.33: "Drive Roll." Hobart's drive-roll technical article explains V-groove and U-groove selection for smooth feeding. (Hobart Brothers)	Hobart supports U-groove rolls for aluminum/soft wire; ESAB also warns that sharp-edged rolls can shave aluminum wire. (Hobart Brothers)
WF-13	Pressure arm	Assembly	Apply controllable force between the wire and the drive rolls.	Pressure-arm body, springs, spring shaft, bushing, washers, pressure knob.	Stainless steel or plated steel arm; spring steel springs; Buna washer in conventional systems; vacuum version should replace unqualified elastomers.	Miller S-74S/S-74D drive-assembly parts list; p.32: "PRESSURE ARM."	Miller lists steel washers and Buna washer in this assembly; for lunar/vacuum, elastomers require replacement or qualification.
WF-14	Pressure adjustment knob	Single part	Let the operator or actuator set drive-roll pressure.	Knob body, threaded insert or extension.	Plastic knob in conventional feeder; aluminum or stainless knob for vacuum/lunar version.	Miller S-74S/S-74D drive-assembly parts list; p.32: "KNOB, w/extension short pressure arm" and "KNOB, w/extension long pressure arm."	Miller identifies the knob but does not define a lunar/vacuum material. Replace unqualified plastic if exposed to vacuum or high temperature.
WF-15	Pressure spring	Single part	Provide spring force for drive-roll pressure.	Compression spring.	Spring steel or stainless spring steel.	Miller S-74S/S-74D drive-assembly parts list; p.32: "SPRING, cprsn" and "SPRING."	Material is an engineering selection; source confirms the part type.
WF-16	Inlet wire guide	Single part	Guide wire into the drive-roll area.	Inlet guide bushing/tube.	Stainless steel, brass, ceramic, or qualified polymer; nylon guides are common in conventional feeders for aluminum.	Miller S-74S/S-74D manual; p.17: "Install wire guides." Miller drive-roll/wire-guide table; p.33: "Inlet Guide."	Miller's drive-roll kit table confirms inlet guides and includes guide kits; material is application-specific.
WF-17	Intermediate wire guide	Single part	Keep wire aligned between roll sets or between inlet and outlet regions.	Intermediate guide bushing/tube.	Stainless steel, brass, ceramic, or qualified polymer.	Miller drive-roll/wire-guide kit table; p.33: "Intermediate Guide."	Miller confirms the part; final material depends on wire alloy, wear, friction, temperature, and

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							vacuum compatibility.
WF-18	Outlet wire guide	Single part	Guide wire out of the drive-roll assembly toward the liner/nozzle path.	Outlet guide body.	Stainless steel, brass, ceramic, or qualified polymer.	Miller threading procedure; p.17: "outlet wire guide."	Material is not fixed by the source; select by wear, friction, and vacuum compatibility.
WF-19	Anti-wear guide	Single part	Protect the wire path near the drive rolls from wear.	Anti-wear guide insert.	Hardened steel, ceramic, or tungsten-carbide insert.	Miller threading procedure; p.17: "anti-wear guide." Miller kit table also says each kit contains an anti-wear guide.	Material is engineering-selected; exact material needs a supplier datasheet.
WF-20	Feed liner	Single part	Guide wire through the downstream feed path after the drive rolls.	Replaceable liner tube.	Steel liner for hard wire; nylon/Teflon-type liner for aluminum in terrestrial systems; for lunar/vacuum, use verified low-outgassing polymer, ceramic tube, or metal guide tube.	Bernard/Tregaskiss's GMAW guide explains the wire path from spool through liner at controlled rate. (Bernard and Tregaskiss) Bernard/Tregaskiss liner article states that the liner guides welding wire from the feeder through the gun cable. (Bernard and Tregaskiss)	Material must be verified against vacuum, temperature, friction, and wire alloy. Conventional GMAW liner materials are not automatically acceptable for lunar vacuum.
WF-21	EBF wire guide tip	Single part	Precisely deliver wire into the electron-beam melt pool.	Guide-tip body with bore.	316L stainless steel, alumina ceramic, zirconia ceramic; copper alloy only if the tip is heated or current-carrying.	NASA EBF ³ space-environment paper states that wire is fed into a molten pool created by an electron beam; p.2.	Material is EBF engineering selection; final material needs thermal, vacuum, and spatter/metal-vapor testing.
WF-22	Gun/feeder adapter	Single part	Mechanically connect the feeder output to the downstream uide/nozzle or gun-side interface.	Adapter body.	Stainless steel or aluminum alloy; ceramic insert if electrical isolation is required.	Miller drive-assembly parts list; p.32: "ADAPTER, gun/feeder." NASA EBF ³ paper states that the wire feeder is attached to the electron beam gun; p.3.	Material is engineering selection based on stiffness, thermal expansion, electrical isolation, and vacuum compatibility.
WF-23	Digital tachometer / feed-rate encoder	Assembly	Measure or infer actual wire feed speed.	Tachometer circuit, encoder disc, connector, signal cable.	Optical encoder disc, PCB, stainless/aluminum bracket, shielded signal cable.	Miller drive-assembly parts list; p.32: "CIRCUIT CARD, digital tach" and "OPTICAL ENCODER DISC." NASA patent also identifies wire-feed-rate sensing.	Material is supplier-specific; final encoder materials need vacuum/radiation-compatible datasheets.
WF-24	Wire feeder base	Single part	Provide the structural base for the feeder.	Base plate.	Aluminum alloy, 304/316L stainless steel, or titanium alloy for lightweight lunar version.	Miller main-assembly parts list; p.28: "BASE."	Material is not specified by Miller; lunar version should select by mass, stiffness, thermal distortion, and vacuum compatibility.
WF-25	Base stiffener	Single part	Increase stiffness of the feeder base.	Stiffener plate or rib.	Aluminum alloy, stainless steel, or titanium alloy.	Miller main-assembly parts list; p.28: "STIFFENER, base."	Material is engineering-selected.

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WF-26	Drive-roll cover	Single part	Protect moving drive rolls and reduce contamination ingress.	Cover body, warning label, fasteners.	Aluminum or stainless sheet; transparent cover only if vacuum and impact compatible.	Miller drive-assembly parts list; p.32: "COVER, drive roll."	Material is not specified by Miller; lunar/vacuum material should be selected by contamination, thermal, and maintenance constraints.
WF-27	Mounting bracket to fixed electron beam gun	Assembly	Mount the feeder relative to the fixed electron beam gun and preserve wire-to-beam alignment.	Bracket, dowel pins, screws, slotted adjustment holes.	Aluminum alloy, 316L stainless steel, titanium; Invar for precision datum features if needed.	NASA EBF ³ space-environment paper states the wire feeder is attached to the electron beam gun; p.3.	Material is engineering-selected based on stiffness, thermal expansion, mass, and vacuum compatibility.